



State-of-Practice of Low-Impact Frogs in Rail Transit

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B O S T O N

Outline

- Introduction to frogs and low-impact designs
- Common low-impact frogs in the transit industry
- Low-impact design features that can be incorporated into standard frog designs
- Noise and vibration from low-impact frogs

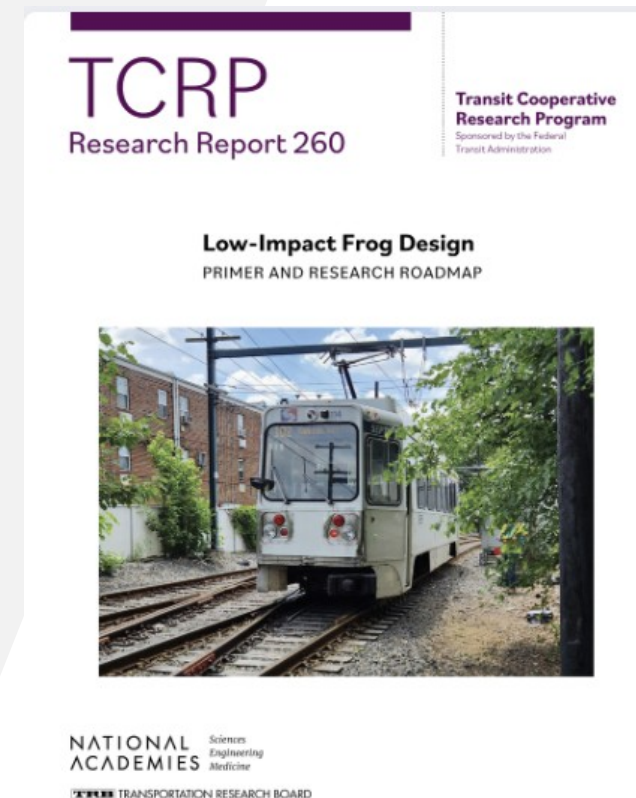


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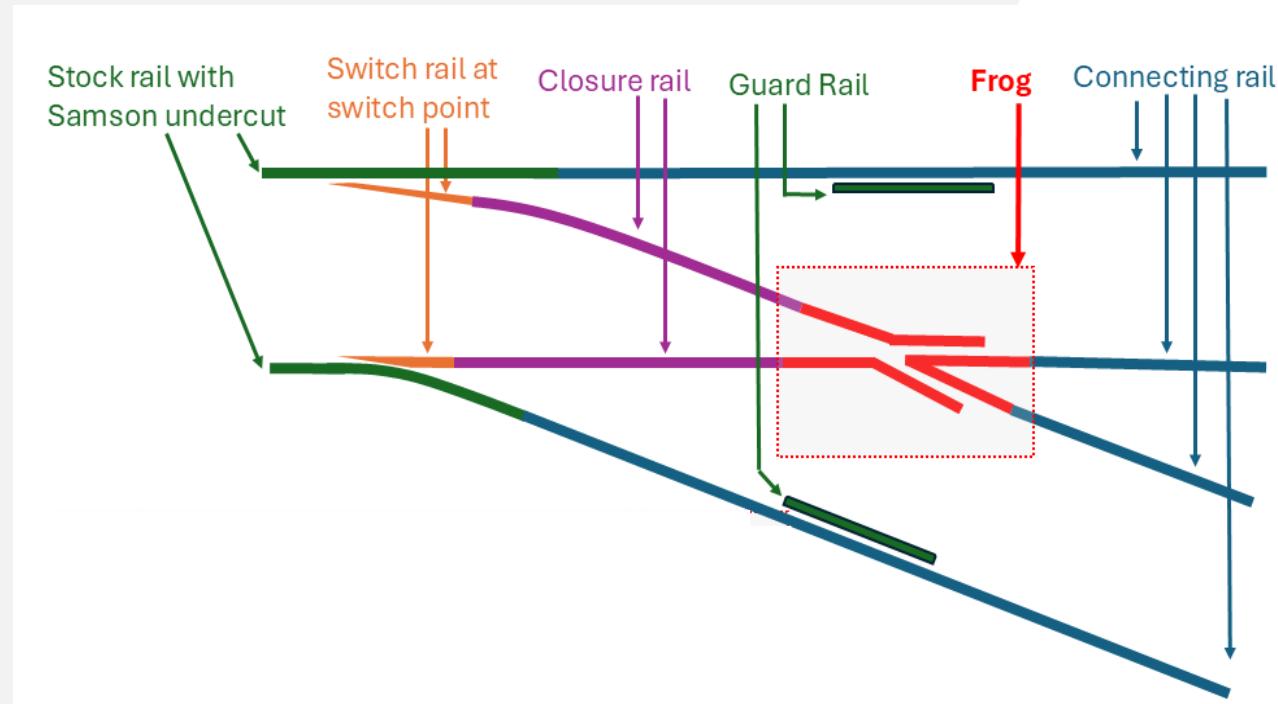
Find the full report:

<https://www.nationalacademies.org/publications/29334>



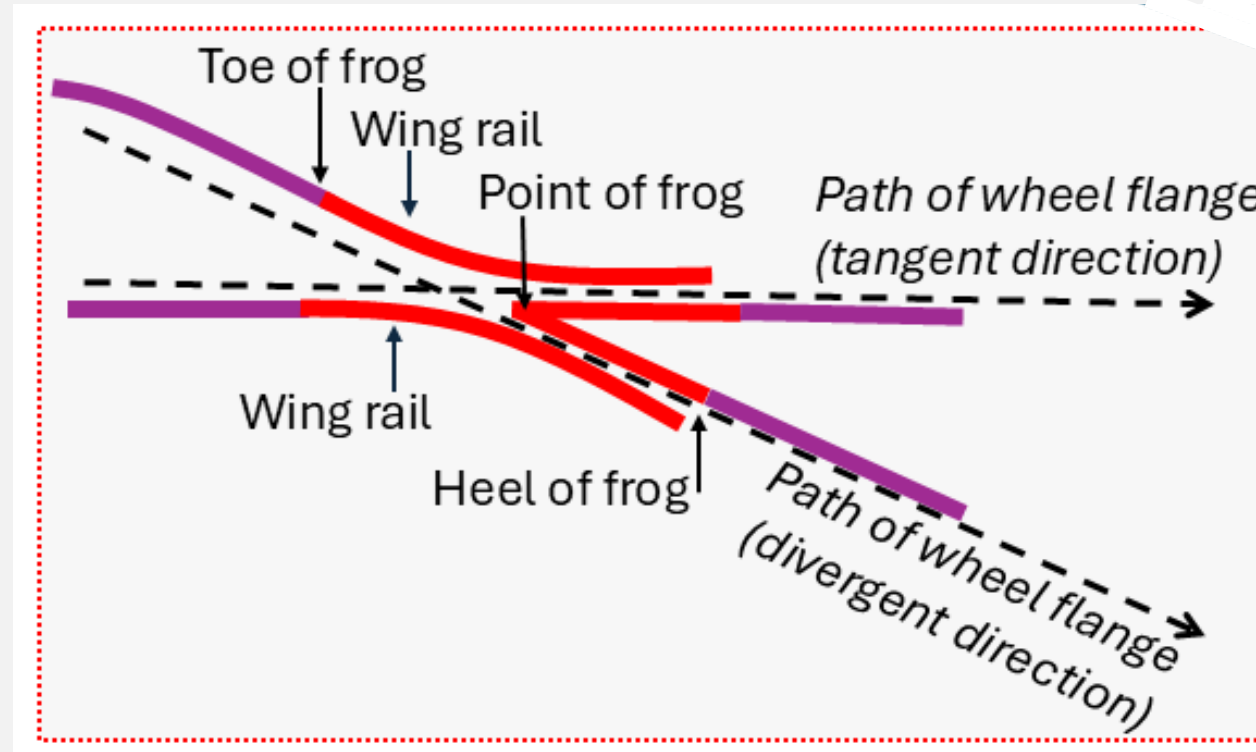
Introduction to Frogs

- Special Trackwork: Track components installed at locations that allow for rails to transfer from one track to another or cross intersecting tracks
- Frog: A component of special trackwork located at a gap in the wheel flangeway where two rails cross
- Typical Turnout:



Introduction to Frogs

- There is a gap in the rail between the wing rail and the point of the frog so the wheel flange can pass through in either direction
- The gap in the rail at the frog is associated with high-impact forces, noise, and vibration even in well-designed frogs



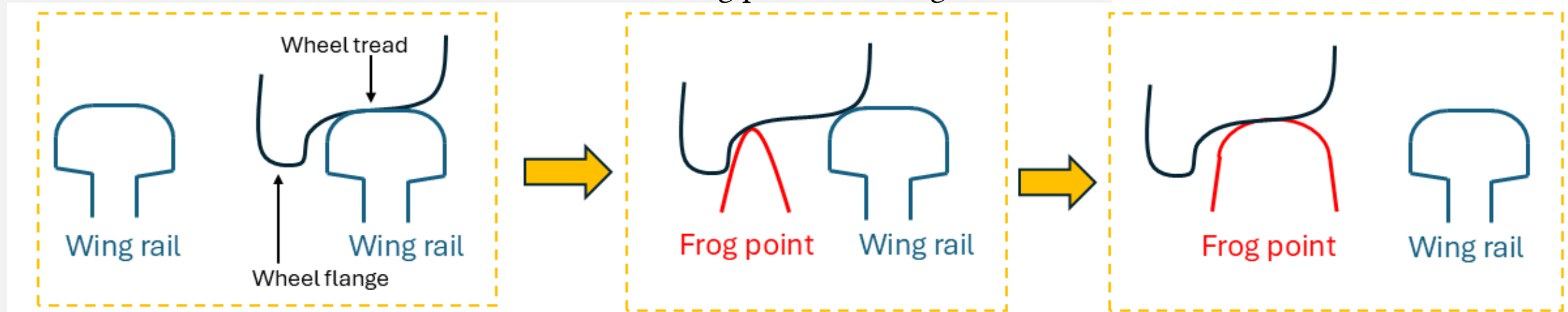
Introduction to Frogs

- A well-designed *wheel-rail interface* through the open flangeway will support the wheel tread as it transfers from the wing rail onto the point of the frog
- Wheel path through the frog:

1. Wheel tread fully supported by wing rail

2. Transition zone where wheel is supported by both the frog point and wing rail

3. Wheel has moved off the wing rail and is fully supported by the frog point



Why low-impact frogs?

- *Optimizing the frog geometry to improve the transfer of the wheel tread from the wing rail over the flangeway gap on the point of the frog can reduce the dynamic forces, lower maintenance costs, improve passengers' comfort, and reduce life-cycle costs*
- A low-impact frog has design elements that are intended to reduce the impact forces at a frog compared to conventional AREMA style RBM frogs



Categories of low-impact frogs

1. Frogs with moveable components so the point is mechanically moved closer to the wing rail to eliminate the gap in the running surface
examples: moveable point, spring-rail
2. Frogs that incorporate ramps that transition the load from the wheel tread to the wheel flange as it passes through the point
examples: flange-bearing, jump or “one-way low-speed”
3. Frogs with low-impact design features that have improvements in geometry or manufacturing compared to conventional RBM frog design, but do not include any additional components
examples: conformal profiles, welded designs, low-impact heel designs, and resilient materials in the track structure

When and where does it make sense to use these low-impact frog designs?





Important Considerations for Frog Selection

- Vehicle / wheel compatibility
- Speed
- Footprint / ROW space
- Noise and vibration benefit
- Cost
- Maintenance needs



	Speed	Footprint / ROW space	Noise and Vibration	Cost	Maintenance	Suggested Applications
RBM	Speed restrictions for diverging move based on frog geometry	Footprint depends on the angle of the turnout: sharp angle turnouts with low-number frogs can be designed to fit small spaces	Excess vibration of 10 dB or more and excess noise of 5 dB close to the turnout compared to tangent track	~\$15,000 as a low-end estimate	Welding is needed when frogs begin to wear; regular tightening of loose bolts or fasteners required	Most common frog design in use today and suitable for most applications
Moveable Point	Popular for higher speed turnouts that have a shallow angle but also have a longer flangeway gap	Additional space needed for signaling equipment	Well-maintained moveable point frogs may significantly reduce excess noise and vibration	3x RBM cost plus cost for additional signaling and labor	Additional maintenance required to keep switch and signaling equipment functioning	Best option to maximize speed in both tangent and diverging directions
Spring	No speed restriction for tangent direction; diverging direction speed restriction usually comparable to RBM	Similar to RBM	Recent data show similar excess noise and vibration levels as RBM frogs	2x RBM cost plus additional for point detection box	Difficult for hi-rail vehicle to push open spring frog in diverging direction	Best option where a closed-gap frog is needed but moveable point or jump frogs are not feasible
Flange-Bearing	Speed restrictions based on the slope of the flange-bearing ramp for both tangent and diverging directions	Longer turnouts with more gently sloped ramps are needed for faster speeds	Generally considered to perform better than RBM frogs: excess vibration of 5 decibels and excess noise of 2.5 decibels close to the turnout	1.3x RBM cost	Flange-bearing ramps need to be frequently welded up to successfully transition wheel from tread to flange	Best option when narrow wheel tread cannot support wheel through typical RBM gap
Jump	Full speed for tangent direction; restricted to 10 mph for diverging move	Footprint similar to RBM; higher guardrail may not be compatible with all vehicle types	No excess noise and vibration in tangent direction; slow speeds in diverging direction	0.7x RBM cost	Minimal maintenance if diverging move is rarely used	Best option for infrequently used or emergency turnouts

Note: **Green** shading indicates positive features, **red** shading indicates negative features, and **blue** shading indicates neutral features **B O S T O**

RBM Frogs

- Rail-bound manganese: solid manganese insert that is wrapped with milled sections of rail and then bolted together
- Very reliable because they have no moving parts and they are the most common frog design, making them familiar to all track maintainers
- The standard AREMA-style RBM frog can be adjusted to custom wheel profiles



RBM Frogs

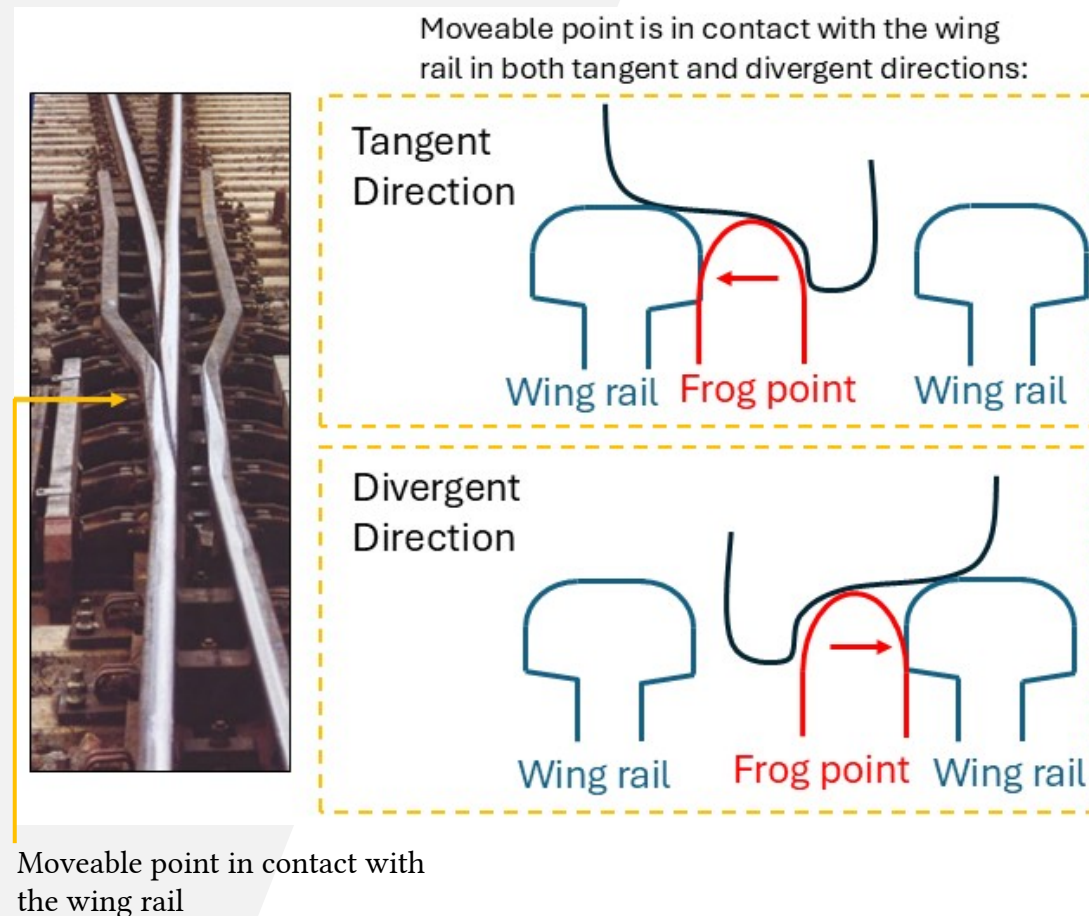
Pros	Cons
Rugged for all loads and conditions	High impact loads generate wear and high noise and vibration levels
Easy to install and maintain	Welding required over time
Standard design makes it easier to procure	Bolts may become loose and need re-tightening

- Agency Interview Takeaways:
 - If its not broke, don't fix it
 - If you are seeing high rates of wear on RBM frogs, custom profiles can help improve performance
 - Maintenance is key to extending the life of the frogs



Moveable Point Frogs

- Gap in the flangeway is eliminated by laterally moving the point of the frog
- Additional signaling cable provides a link between the switch and the frog; additional switch controls are required at the frog
- Allow for safer movement through a wider flangeway gap, especially at higher speeds
- Generally cost 3x more than RBM frogs



Moveable Point Frogs (MPFs)

Pros	Cons
Smooth ride	Greater cost
Greater speed	More frequent inspections
Less wear on wheels	Need for switch machines and more right-of-way space
No flange-way gap	More signal cabling
Lower noise and vibration	Increased training to maintain

- Agency Interview Takeaways:
 - Vancouver Skytrain uses only moveable point frogs due to their particular wheel profile; routine maintenance is needed to minimize corrugation and ensure good ride quality through the frogs
 - Good maintenance needed for switching components and cabling; if MPF are rare on your system increased training is necessary



Spring Frogs

- Gap in the flangeway is eliminated in the tangent direction only
- Most suitable for locations where diverging movements are rare
- Light-weight vehicles, maintenance vehicles, and some wheel profiles may have difficulty pushing open the spring frog
- Generally cost 2x more than RBM frogs

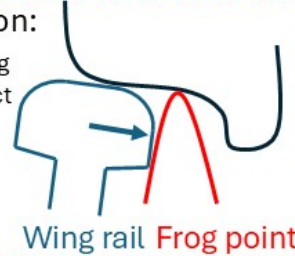
Spring-loaded wing rail
in contact with point



Slide motion when wheel pushes
open wing rail for diverging move

Tangent Direction:

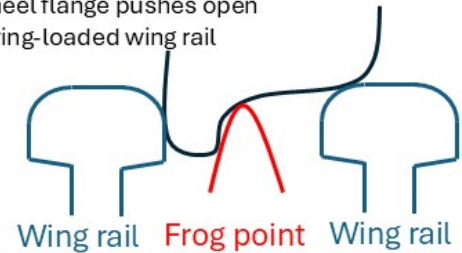
Spring-loaded wing
rail stays in contact
with point
eliminating gap in
flange-way



Wing rail Frog point

Divergent Direction:

Wheel flange pushes open
spring-loaded wing rail



Wing rail Frog point Wing rail



Spring frogs

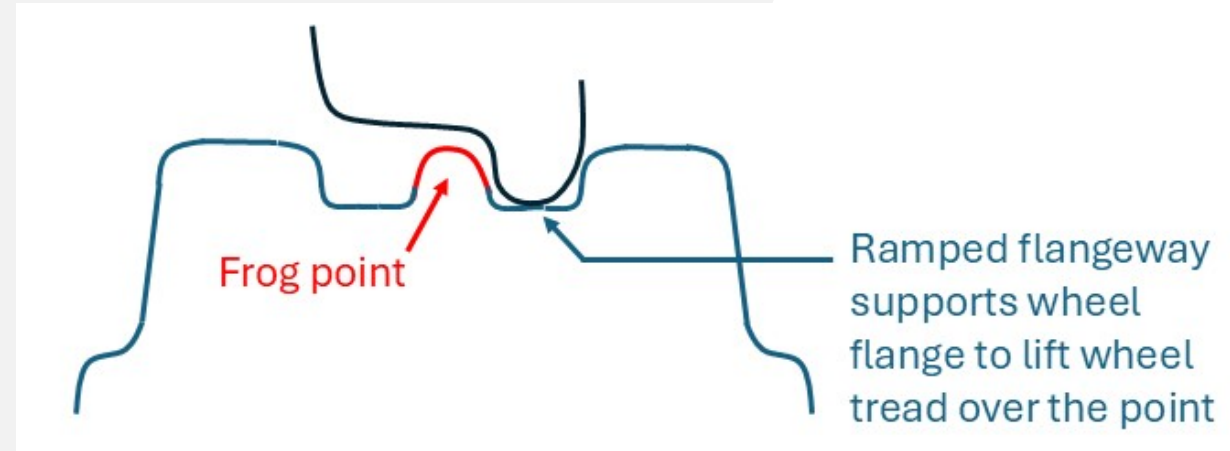
Pros	Cons
Eliminates the gap in the flangeway for the tangent direction without requiring additional switching gear.	Light vehicles and hi-rail vehicles may have difficulty pushing open the closed spring frog when moving through the diverging direction creating reliability and maintenance issues.
Less expensive than moveable point frogs but about twice as expensive as an RBM frog and procurement times may be long.	Recent data shows similar noise and vibration levels compared to conventional RBM frogs

- Agency Interview Takeaways:
 - Reliability is a big concern: light-weight vehicles, systems with multiple vehicles, hollow wheels, and maintenance vehicles may have trouble moving through the spring frog
 - One system is phasing out spring frogs because maintenance vehicles cannot move through the frogs easily
 - Higher speed trains running in the diverging direction may cause a very loud noise when the spring frog “slaps” back against the point



Flange-Bearing Frogs

- Transfer the support from the wheel tread to the wheel flange using a ramp in the special trackwork
- Allowable operational speeds depend on the length of the ramp; higher speeds require a longer ramp with a shallower slope
- Popular in street running systems where the wheel profiles may have narrow tread widths and continuous tread support cannot be provided through the transition zone



Flange-Bearing Frogs

Pros	Cons
May reduce noise and vibration	Ramps need to be consistently welded up to maintain the height necessary to transfer the wheel load from the tread to the flange.
May reduce frog wear by minimizing impact at the point and wing rail.	High operational speeds can require long flange-bearing ramps, which may not be practical for all locations. Shorter ramps require slower speeds.
Compatible with wheel profiles with narrow tread widths.	Large amounts of flange-bearing frogs may require flat to nearly flat wheel flanges to minimize metal flow.

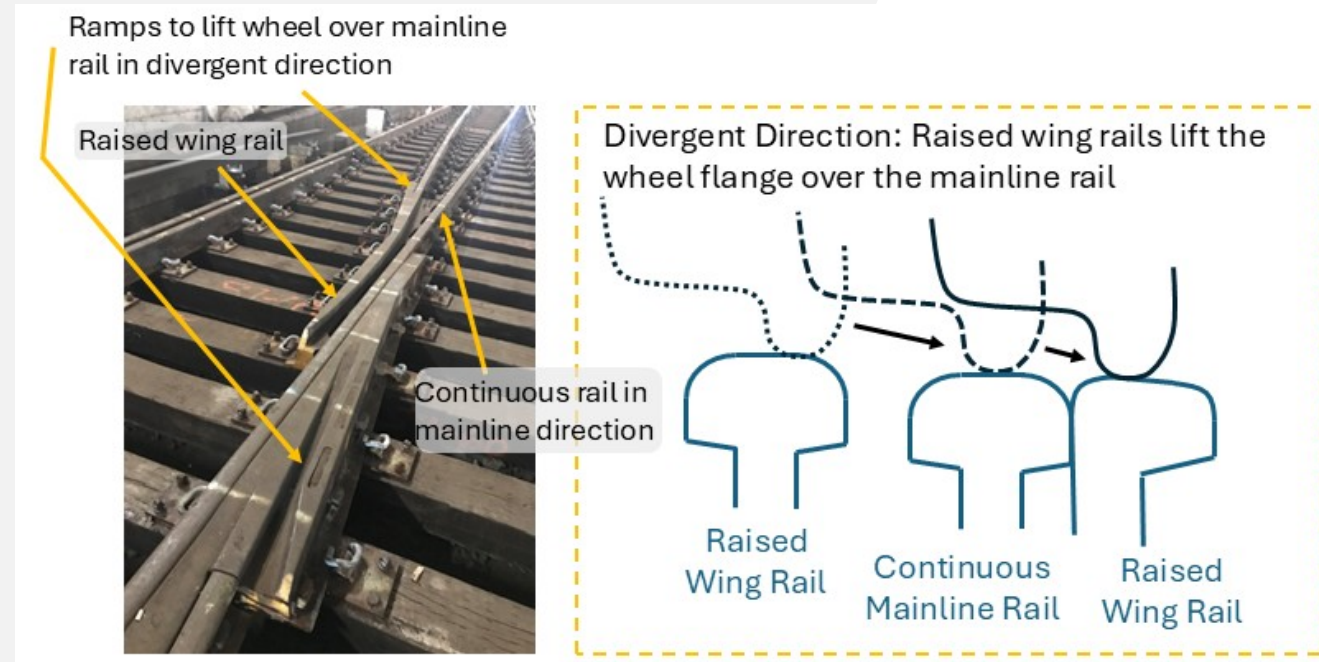
- Agency Interview Takeaways:

- One agency has retrofit RBM frogs to be flange-bearing on its light-rail system as a noise and vibration mitigation measure
- Flange-bearing frogs may not be practical at agencies with short flanges
- Rigorous maintenance of both ramps and wheels is needed
- For street-running systems, maintenance is also needed for spalling concrete and road surface irregularities



Jump, Lift, or OWLS Frogs

- Continuous rail in the tangent direction
- On the diverging route, the flangeway and wing rail are ramped up to allow the wheel to pass over the mainline running rail head and open flangeway
- Slow train speeds are required for the diverging move
- Very effective where diverging move is made infrequently, such as emergency turnouts



Jump, Lift, or OWLS Frogs

Pros	Cons
Continuous rail in the mainline direction with no speed restrictions	Very slow speeds for diverging direction.
Eliminates all excess noise and vibration in the mainline direction	If the diverging direction is frequently used, the flange-way ramps will need to be consistently welded up to maintain the necessary height.
Lower cost compared to RBM frogs and other low-impact frog designs.	Not compatible with all wheel profiles or light-rail vehicle types.

- Agency Interview Takeaways:
 - There may be clearance issues with low-floor vehicles
 - Successful installation on a street-running trolley system where the trolley line “jumps” over the freight tracks; much more durable than a standard transit frog subjected to freight traffic
 - Likely to be the most reliable and longest lasting for emergency turnouts that use the diverging move infrequently, if the agency has a compatible wheel profile and vehicle type



Low-Impact Design Features

- Design features that have improvements in geometry or manufacturing compared to a conventional RBM frog design, but do not include any additional components:

Design Feature	Description
Conformal Profiles	Conformal profile frogs have a slope on the wing rail and point that is designed to match with the wheel shape that runs over them. Conformal profiles need to be designed to specifically match the wheel profile of the agency.
Welded Designs	In welded designs, the frogs are welded to the running rail instead of using bolted joints. This results in a smoother running surface as well as reduced maintenance needs due to not using bolts which can rattle loose.
Solid Manganese Designs	Made of solid manganese steel instead of the manganese insert used in conventional RBM frogs. These frogs eliminate the impact forces where the wing rail steel mates with the manganese outline. Because manganese steel is highly resistant to wear and impacts, these frogs are very durable.



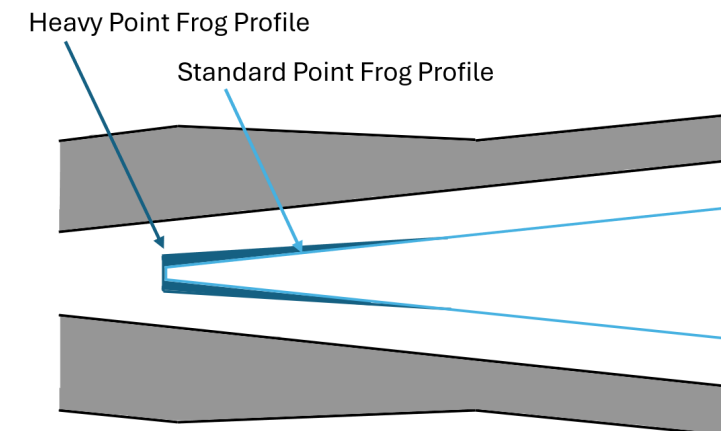
Low-Impact Design Features:

Design Feature	Description
Tapered Heel Designs	Conventional frog heels typically have a 90 or 60 degree angle between the frog and running rail. The tapered heel has a much sharper angle that results in a much larger contact area between the frog and the running rail and a lower dynamic load.
Heavy Point Designs	Heavy point frog designs widen the frog point. The wider frog point is capable of carrying heavy axle loads without excess deformation. Due to the narrower flangeway, speed restrictions may apply.
Resilient Materials in Track Structure	Track stiffness influences the impact load at special trackwork. Reducing the track stiffness by incorporating resilient materials into the track structure can reduce the impact loads. Examples of resilient materials are under tie pads, ballast mats, resilient fasteners, and rail boots. These materials may also reduce vibration levels if they are designed to be effective at lower frequencies.

Conventional Heel:



Tapered Heel:

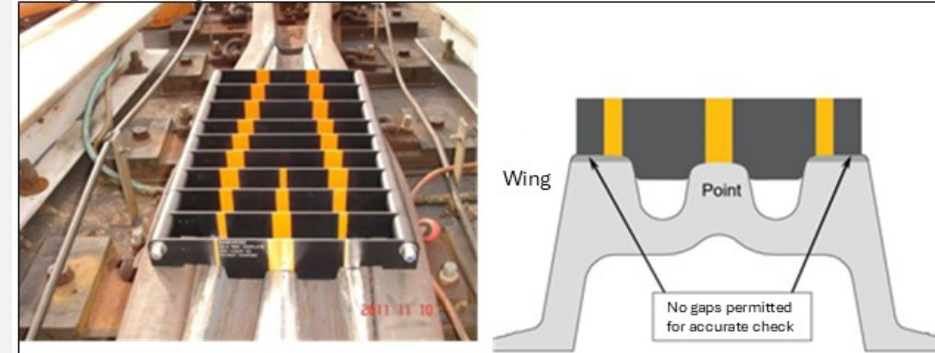




Most Frequent Comment from Agencies: Maintenance is Key!

- Inspection: using special jigs can help to assess wear and plan for maintenance
- Grinding: Regular grinding is needed to control corrugation and maintain a good wheel/rail profile; hand grinding around frogs is a specialty skill
- Welding: needed to maintain the frog point, frog heel, and flangeways in a state of good repair; manganese requires specialty welders
- Wheel Truing: Wheels also need to be in good condition
- Workforce development: good maintenance requires a well-trained workforce

Inspection jig:



Welding jig:



Source: Skoblenick 2014

Do low-impact frogs generate lower noise and vibration levels?

- Current FTA guidance on noise and vibration for standard RBM frogs:
 - Increase Vibration levels by +10 dB
 - Increase Noise levels by +5 dB
- Generally expected that low-impact frogs reduce noise and vibration levels, but there is no official FTA guidance and very little measured data available
- Many factors affect noise and vibration performance at frogs: age of frog, maintenance practices, vehicle type, speed, wheel condition, etc.
- The TCRP project funded detailed noise and vibration measurements at frogs on two systems:
 - LA Metro light-rail: RBM, spring, and WBM frogs
 - SEPTA light-rail and trolley: RBM, flange-bearing, and jump





Noise Measurement Results: RBM and Spring Frogs on Ballasted Track

- Excess Noise = Noise at Frog – Noise at Tangent Track

Site	Excess Noise at 12 ft	Excess Noise at Residence
LA Metro RBM, 30-40 mph	7 dB	2 dB at 32 ft
LA Metro Spring, 30-40 mph	2-7 dB	0-3 dB at 31 ft
Septa RBM #8, 15-20 mph	2-4 dB	0-2 dB at 75 ft
Septa RBM #6, 15-20 mph	6-8 dB	0 dB at 77 ft

Variation due to speed

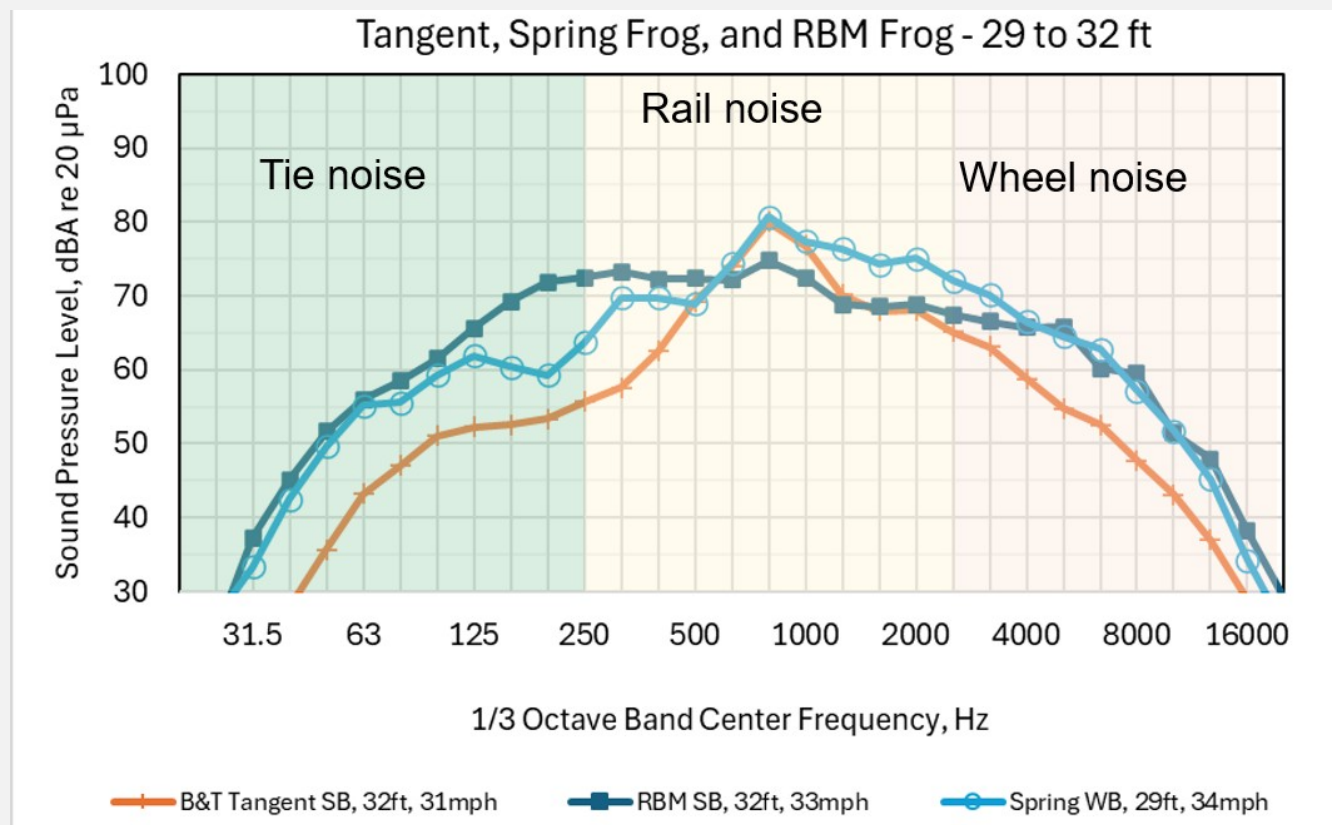
Variation due to wheel condition

Variation due to terrain differences

- Conclusions:
 - Excess noise from the frog reduces with distance
 - FTA guidance is likely overpredicting frog noise at residences (30 ft + from tracks)
 - Excess noise was similar for both systems with varying vehicle types and maintenance practices
 - Excess noise from spring frog and RBM frog are similar



Noise from Tangent Track vs. RBM Frog vs. Spring Frog



- Excess noise from frogs is greatest at low and high frequencies, where noise is radiated from ties and wheels
- Excess noise from ties and wheels may be perceptible, but may not contribute much to overall noise levels



Noise Measurement Results: WBM and Flange-Bearing Frogs on Embedded Track

- $\text{Excess Noise} = \text{Noise at Frog} - \text{Noise at Tangent Track}$

Site	Excess Noise (SEL)
LA Metro WBM* tangent, 19 mph	<0 dB at 26 ft
LA Metro WBM* diverting, 10 mph	1-7 dB at 39 ft
SEPTA flange-bearing in depot, <10mph	4 dB at 36 ft
SEPTA Flange-bearing (high), <10mph	7 dB at 29 ft
SEPTA Flange-bearing (typical), <10 mph	3 dB at 29 ft

Variation due to speed

High levels likely due to wheel condition

* WBM is a welded, boltless manganese frog, which is a low-impact design with a fixed point

- **Conclusions:**
 - Higher excess noise when using the diverting direction
 - For SEPTA flange-bearing frogs, poor condition of embedded track (spalling and chipping) likely contribute to higher noise levels
 - LA Metro WBM tangent frog may be performing better than the RBM and spring frogs, but it is difficult to make comparisons over different track types and different speeds

SEPTA:



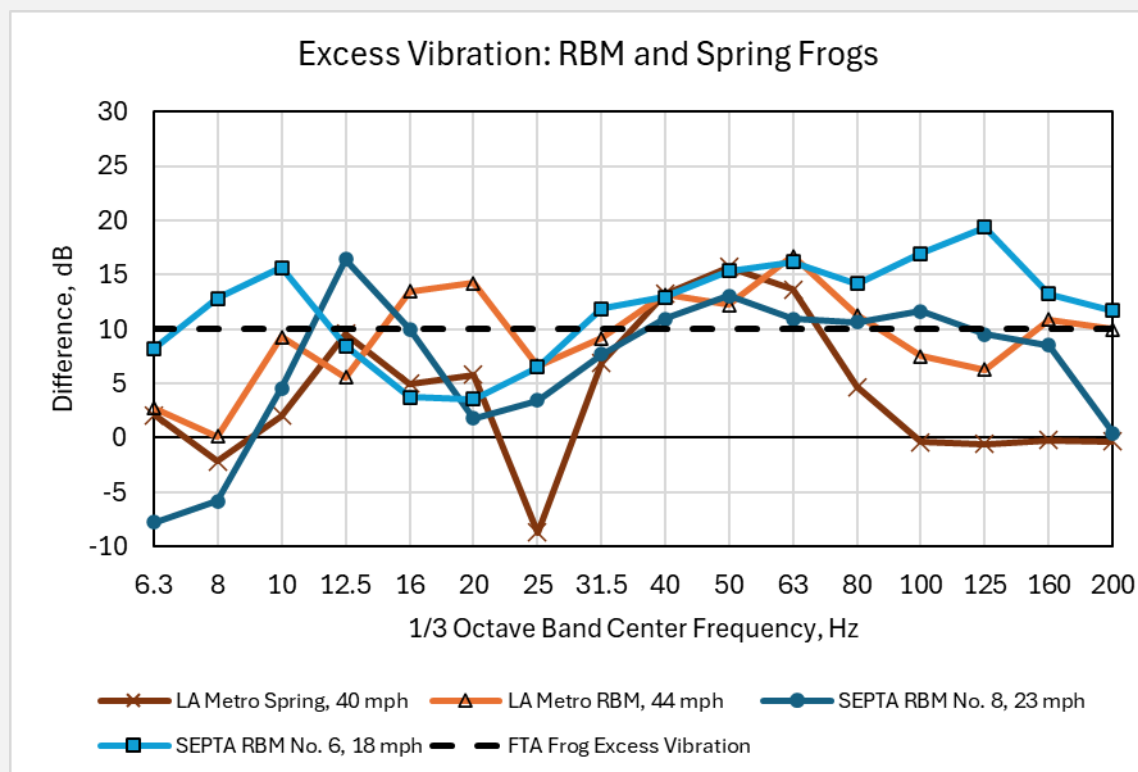
LA Metro:



Vibration Measurement Results: RBM and Spring Frogs on Ballasted Track

- Excess Vibration = FDL^* at Frog – FDL^* at Tangent Track

* Force Density Level

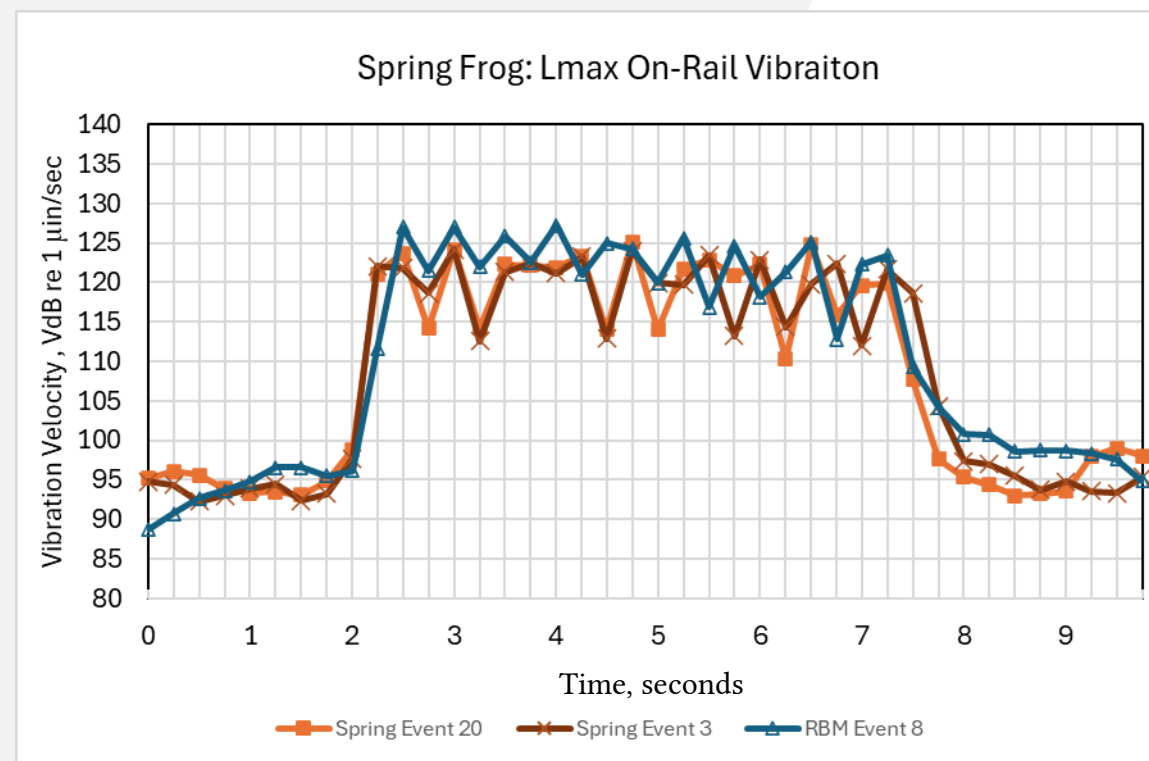
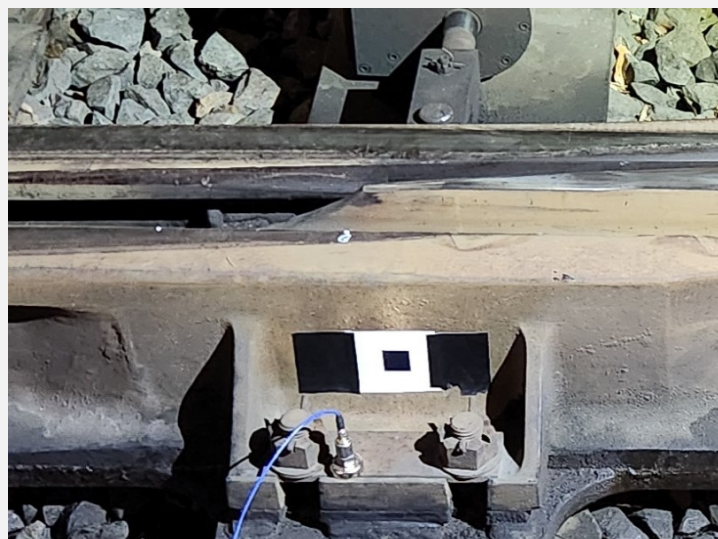


Conclusions:

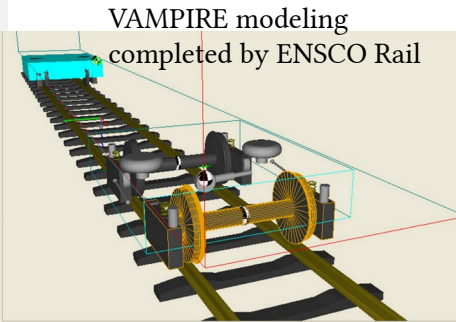
- FTA guidance may be underpredicting frog vibration
- Excess vibration was similar for both systems with varying vehicle types and maintenance practices
- Spring frog shows comparable levels at mid-frequencies, and lower levels at high frequencies

Vibration Measurement Results: RBM and Spring Frogs on Ballasted Track

- Vibration measured on the spring frog is similar to the vibration measured on the RBM frog



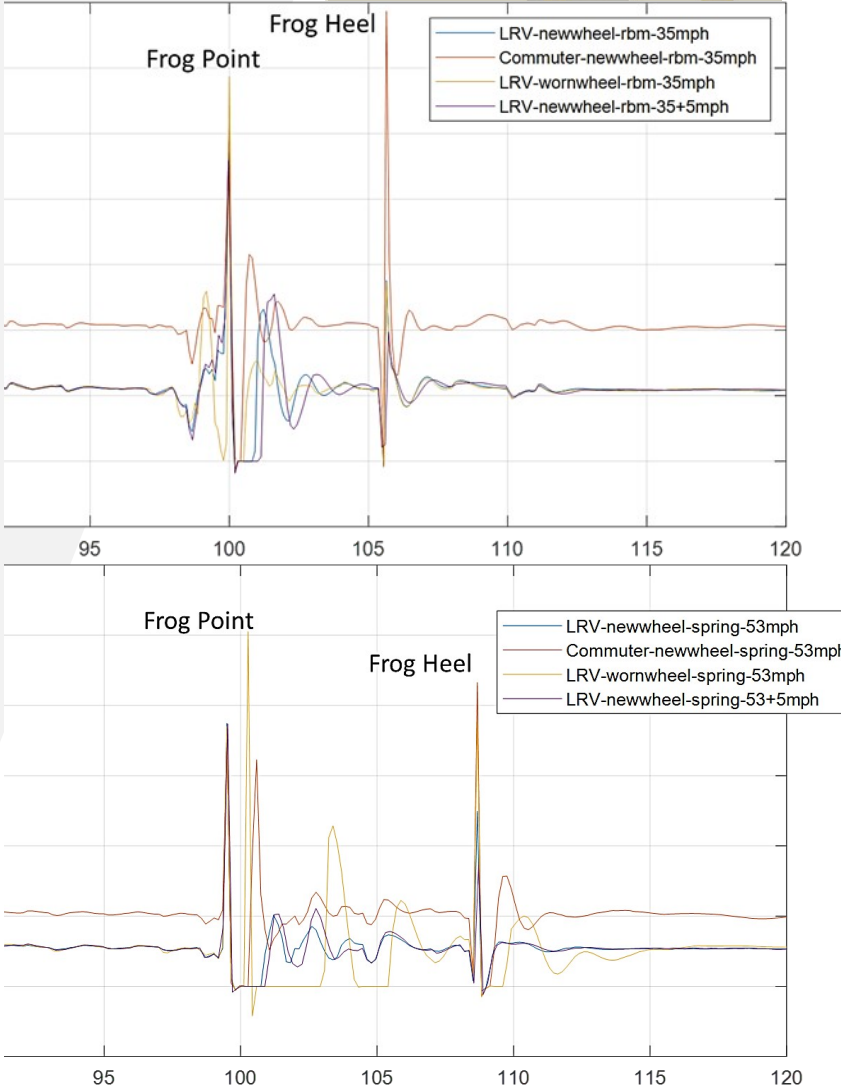
VAMPIRE Modeling Results: RBM and Spring Frogs on Ballasted Track



- VAMPIRE model constructed with miniprof data from field sites and a comparable vehicle
- Spring frog does not show lower force levels compared to the RBM frog:

Condition	Frog Point Peak Transient Vertical Wheel Force (lbf)	Frog Heel Peak Transient Vertical Wheel Force (lbf)
RBM Frog, LRV, new wheel, 53 mph	53,000*	17,000
Spring Frog, LRV, new wheel, 53 mph	74,900*	49,900
RBM Frog, LRV, worn wheel, 35 mph	58,700	27,300
Spring Frog, LRV worn wheel, 53 mph	101,000	75,800

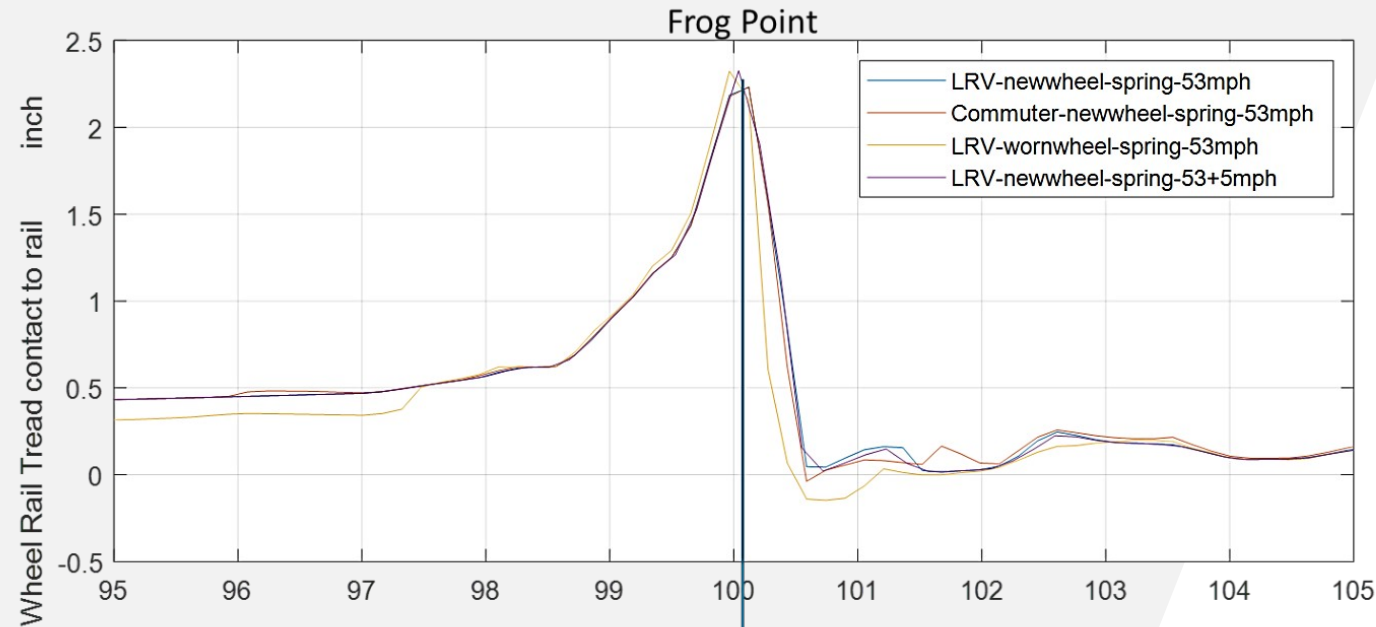
*40% force increase for spring frog vs RBM frog is ~1.5 dB change



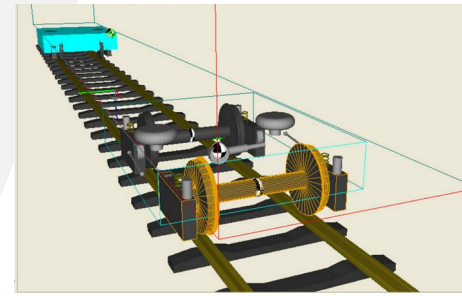


VAMPIRE Modeling Results: RBM and Spring Frogs on Ballasted Track

- VAMPIRE modeling was able to show lateral shift of contact patch:



B O S T O

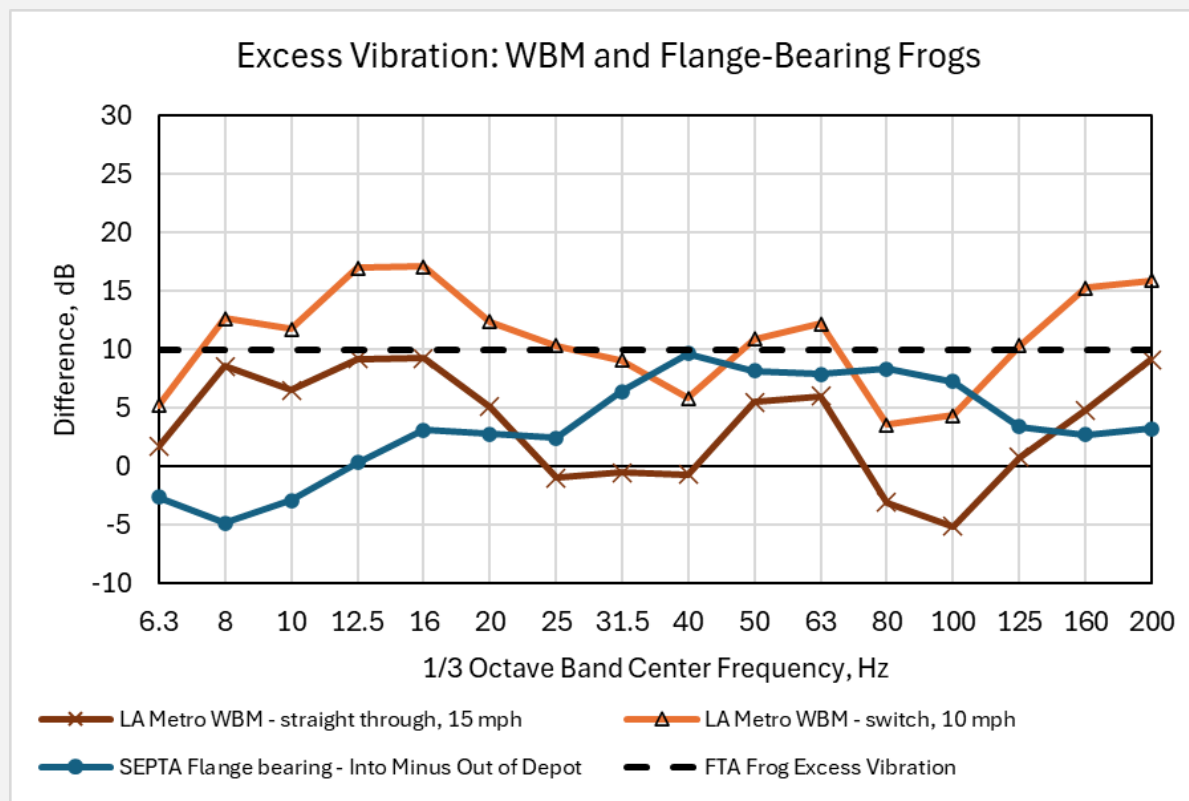


VAMPIRE modeling completed by ENSCO Rail

Vibration Measurement Results: WBM and Flange-bearing Frogs on Embedded Track

- Excess Vibration = FDL^* at Frog – FDL^* at Tangent Track

* Force Density Level



Note: No tangent track reference data available for SEPTA embedded track

Conclusions:

- Excess vibration is higher for the switching movement
- Additional data from WBM frog at higher speeds is needed to confirm if it achieves lower vibration levels
- SEPTA excess vibration levels may be due to differences in speed as trains enter versus exit the depot

Do low-impact frogs generate lower noise and vibration levels?

- Many confounding variables:
 - Controlled test sites with consistent speeds are needed to better answer this question
 - WBM frogs showed better performance than the Spring frog, but WBM data was only collected at relatively low speeds
- Vibration mitigation at frogs may be better achieved by adding resilient materials (ie. ballast mat or resilient fasteners) than by selecting a moveable-point or spring frog.
- FTA guidance may be over-predicting the noise contribution from frogs



Conclusions

- Frog designs must be carefully selected to be compatible with vehicle and wheel profile and to meet speed requirements
- Good maintenance of frogs and wheels is key to extend frog life and reduce impact forces, noise, and vibration levels
- Specialty low-impact frogs don't always reduce noise, vibration, and impact forces
- Some low-impact design features show promise for transit applications: conformal designs, welded designs, and resilient materials

